MIL-T-2283E
20 November 1969
SUPERSEDING
MIL-T-2283D
16 February 1966

MILITARY SPECIFICATION

TAPE, TEXTILE, NYLON, WOVEN, WHITE OR DYED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of woven nylon tape.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-43334 - Webbing and Tape, Textile, Packaging and Packing of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8315

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2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580)

3. REQUIREMENTS

3.1 Standard sample.- The tape shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.-

3.2.1 Yarn.- The nylon yarn used in the manufacture of the tape shall be 210 denier ± 5 percent, bright, high tenacity, polyamide prepared from hexamethylene diamine and adipic acid, or its derivatives when tested as specified in 4.2.1.

3.3 Color.- The tape shall be natural (white) or dyed as specified (see 6.2).

3.3.1 Color matching.- The dyed tape shall match the standard sample under natural (north sky) daylight, or artificial daylight having a color temperature of 7500° Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness.- The dyed tape shall show fastness to wet dry cleaning, laundering, and light equal to or better than the standard sample when tested as specified in 4.3. When a standard sample of reference is not available, the dyed tape shall show good fastness to wet dry cleaning, laundering and light when tested as specified in 4.3.

3.4 Physical requirements.- The finished tape shall conform to the requirements shown in table I when tested as specified in 4.3.

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TABLE I.- Physical requirements

Width tolerance + 1/16 - 1/32	Texture		Weight per gross yds. (min.)	Breaking strength (min.) (warp)
	Total ends (min.)	Picks per inch (min.)		
Inches			Ounces	Pounds
1/2	65	40	9.00	90
5/8	79	40	11.00	140
3/4	95	40	13.50	190
1	127	40	17.00	240
1-1/4	157	40	22.00	265
1-1/2	187	40	26.00	290

* 3.4.1 Weave.- The tape shall have a plain weave with one filling yarn per shed and woven selvages on both edges.

3.4.2 Width.- The width shall be as specified (see 6.2).

* 3.5 Nonfibrous materials.- The water-soluble and chloroform-soluble materials shall not exceed 2.0 percent when tested as specified in 4.3.

* 3.6 pH.- The pH value of the water extract of the dyed and finished tape shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.3.

* 3.7 Length.- Unless otherwise specified, the tape shall be furnished on standard commercial spools, with a minimum gross length of 288 yards per spool. Each spool shall contain not more than five pieces and unless otherwise specified, the minimum length of any piece shall be 10 yards (see 6.2).

* 3.8 Fiber identification.- Each spool of tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

* 3.9 Identification label.- Each spool of tape shall have an identification label attached to the head end of the spool in accordance with MIL-W-43334.

* 3.10 Workmanship.- The tape shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item.- Defects found during the examination shall be classified in accordance with 4.2.2.1, through 4.2.2.3.

4.2.2.1 Yard-by-yard examination.- The required yardage of each spool shall be inspected on both sides and visual defects classified in accordance with table II. All defects found shall be counted regardless of their proximity one to another, except where two or more defects represent a single local condition of the tape, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be 1 linear yard. The sample size shall be in accordance with level II of MIL-STD-105. The acceptable quality level shall be 1.5 defects per 100 units (yards). The lot size shall be expressed in units of 1 yard. An approximately equal number of yards shall be examined from each spool selected. The number of spools from which the sample is to be selected shall be in accordance with table IV of this specification.

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TABLE II.- Defects

Broken or missing yarns	Two or more regardless of length or single more than 6 inches in length (warp) or full width (filling).
Coarse or heavy yarn	Three or more times the thickness of normal yarn.
Crease or wrinkle	Permanently embedded in edge or body.
Edge	Cut, torn, frayed, puckered, rolled, loose, scalloped or otherwise poorly constructed.
Filling	Mixed or two or more tight or fine picks.
Floats or loops	Multiple
Cut, hole, tear or smash	Any
Knots	Two or more; or untrimmed ends extending from surface of tape 1/
Misweave	Two or more yarns (ends or picks) incorrectly interlaced
Spot, stain, streak or shade bar	Any 1/
Weak or open place or abraided area	Any 1/
Wavy, ridgy, rolling or puckered tape	Regardless of cause and will not flatten under manual pressure
Width	Not within specified tolerance

1/ Clearly visible at normal inspection distance (approximately 3 feet).

4.2.2.2 Overall examination.- Each defect listed in table III shall be counted not more than once in each spool examined. The sample unit for this examination shall be one spool. The sample size and acceptance number shall be as shown in table IV.

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TABLE III.- Overall examination

Overall uncleanness.
Uneven weaving or construction throughout.
Off shade (not within established tolerances).
Color other than specified.
Mottled, cloudy, or streaky throughout.
Not labeled in accordance with Textile Fiber Products Identification Act.

4.2.2.3 Length examination.-

4.2.2.3.1 Examination for length of individual spool.- Each spool in the sample shall be examined for the defects listed below. The sample unit for this examination shall be one spool. The sample size and acceptance number shall be as shown in table IV. The lot size shall be expressed in units of 1 yard each.

Defects
Gross length of any spool more than 2 yards less than length marked on label.
Length of spool less than specified.
Length of individual piece of tape less than minimum specified.
Pieces per spool: - More than five pieces - More or less than number of pieces indicated by supplier's marking on label.

4.2.2.3.2 Examination for total yardage in sample.- The lot shall be unacceptable if the total of the actual gross lengths of spools in the sample is less than the total of the gross lengths marked on the label.

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TABLE IV.- Sample size

Lot size in yards	Sample size in spools	Maximum number of defects acceptable in sample 2/
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than 3 spools, each spool in the lot shall be examined.

2/ Except that the acceptance number shall be zero for color and uniformity of shade defects found in overall examination 4.2.2.2.

4.2.3 Examination of preparation for delivery requirements.- An examination shall be made in accordance with the provisions of MIL-W-43334 to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with.

4.3 Testing of the end item.- The methods of testing specified in FED-STD-191 wherever applicable and as listed in table V shall be followed. The physical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for test purposes shall be 6 linear yards of tape. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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* TABLE V.- Test methods

Characteristic	Requirement paragraph	Test method
Nylon yarn	3.2.1	1/
Texture		
Total ends warp	3.4	5050 2/
Picks per inch	3.4	5050 2/
Weight oz./gross yards	3.4	4.3.1
Breaking strength	3.4	5100 3/
Weave	3.4.1	Visual 4/
Colorfastness to:		
Wet dry cleaning	3.3.2	5622
Laundering	3.3.2	5614
Light	3.3.2	5660
Nonfibrous material	3.5	2611 1/
pH	3.6	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and result reported as "actual count".

3/ Breaking strength shall be determined on the full width of the specimen and the jaw dimensions shall be greater than the width of the specimen.

4/ One determination per sample unit and results reported as "pass" or "fail".

4.3.1 Weight per gross yards.- Weigh six yards of tape in a relaxed condition (no tension) on a balance or scale, accurate to 0.01 ounce and multiply by 24 to obtain the weight of 144 yards. There shall be one determination per sample unit.

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5. PREPARATION FOR DELIVERY

5.1 **Packaging.-** Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A, B and C.- Tape, put up as specified, shall be packaged in accordance with the applicable requirements of MIL-W-43334.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- Tape shall be packed in accordance with the applicable requirements of MIL-W-43334.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-W-43334.

6. NOTES

6.1 Intended use.- The tape covered by this specification is used in the fabrication of clothing and individual equipment.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Color required (see 3.3).
c. Width required (see 3.4.2).
d. Length of spool required, if other than specified (see 3.7).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample.- For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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